VLM-Verified Counterfactual Hindsight for Sparse-Reward Manipulation*

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Extended Abstract

Problem. Sparse-reward goal-conditioned manipulation is a temporal credit-assignment problem: in Gymnasium-Robotics Fetch [3] (Push / PickAndPlace / Slide), a randomly-initialized policy succeeds in $< 1\%$ of episodes, and the lone terminal reward bit at the end of a 50-step trajectory decays exponentially through temporal-difference (TD) bootstrapping. Hindsight Experience Replay (HER) relabels failed trajectories with achieved goals but provides no signal about *which* timestep caused failure; standard prioritized experience replay (PER) cannot recover this in sparse regimes because the TD error is itself the bottleneck. Concurrent VLM-RB [22] uses frozen vision-language models (VLMs) as an *additive* priority bias; we argue the *multiplicative* form is the principled choice and pair it with a simulator-verified counterfactual channel.

Methodological contributions. **(1) Semantic PER as importance-sampled posterior reweighting:** we recast VLM-guided replay as a trajectory-conditional posterior $q_\phi(t^\star \mid \tau)$ over the failure-causing timestep, and show that multiplying the standard TD-error priority by q_ϕ admits a clean importance-sampling (IS) correction with an explicit bias bound: VLM miscalibration inflates sampling *variance* but never corrupts the value target; the additive VLM-RB mixture implicitly retargets the Bellman loss. **(2) VLM-Verified Counterfactual Hindsight:** we close the dominant failure mode of language-only counterfactuals (100% teleport-collapse of Opus 4.7 on FetchPush) by asking the VLM for a *corrective action sequence* rather than a hindsight goal, executing it in a simulator fork, and admitting only physics-consistent, reward-positive relabels into the buffer (confidence 1.0, zero modeling error).

Headline findings. (i) **Verified-CF (counterfactual) matches VLM-CF, ties on Slide.** At 500k soft actor-critic (SAC) + HER + verified-CF updates the verified-CF agent reaches mean success rate 0.606 ($\pm$ standard error, SE) across the three Fetch envs ($0.85/0.35/0.617$ on Push / PickAndPlace / Slide), tying VLM-CF (0.622 mean, $\Delta = -0.016$) and reaching a higher mean on FetchSlide (0.617 vs. 0.550; $+0.067$, within seed variance at $n = 3$). Both methods gain $+0.45/+0.434$ on Slide over PER@3M / HER@250k; Fig. 1. (ii) **Pre-registered kill verdict, honored.** The Phase-1 Oracle-CF study violated the $+0.10$ kill threshold on FetchPickAndPlace at the 250k decision horizon ($\Delta = -0.05$); we honored the kill and pivoted the headline to the IS-posterior framing and the verifier. (iii) **Prompt-architectural failure.** A two-vendor three-model sweep (GPT-4o / Opus 4.7 / Sonnet 4.5) on 6 synthetic and 10 real Fetch failures locates teleport-collapse as prompt-architectural ($0/6$ vs. $4/4$ for `(GPT-4o, achieved_goal)` vs. Opus), not model-specific. (iv) **Cross-task transfer.** A single prompt template yields $12/12$ parse / task-relevance / plausibility on Push / PickAndPlace / Slide.

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Honest negative. Verified-CF has a cold-start regime (PnP seed 42: 100% verifier-rejection over the first ~ 80 calls), reducing to a costlier HER. The multiplicative-vs-additive distinction is analytical (Eq. 1); the Sharony VLM-RB head-to-head is pre-staged, camera-ready.

1 Introduction

The core difficulty of sparse-reward manipulation is temporal credit assignment: with only one reward bit at the end of a 50-step episode, every standard TD-error signal decays exponentially back from the terminal timestep, leaving the bulk of each trajectory effectively invisible to the optimizer. In Gymnasium-Robotics Fetch environments, a randomly-initialized policy succeeds in under 1% of episodes, so the early replay buffer contains almost no positive signal to propagate forward. Prioritized experience replay (PER) [21] sharpens sampling toward high-TD-error transitions, but this sharpening is itself downstream of a credit signal that decays exponentially away from the rare goal-achievement timestep. Hindsight Experience Replay (HER) [1] partially repairs this by relabeling failures as successes for an alternative achieved goal, but the relabel target is constrained to the agent's actually-realized trajectory, so the most instructive transition (the one where failure became inevitable) receives no special treatment.

Recent work uses foundation VLMs as "oracles" inside the RL loop, both for reward shaping [19, 10], replay-buffer prioritization [22], and failure detection in manipulation [5]. We frame these uses as *importance-sampled stochastic optimization with a foundation-model proposal distribution*: the VLM is a frozen approximation to the (intractable) posterior $p(t^\star \mid \tau)$ over which transitions caused an episode's outcome. This framing yields four contributions:

(1) An IS-posterior framing of VLM-guided replay (§3). We re-derive PER inside a non-uniform-replay taxonomy, locating our scheme as *proposal-shaping* with a frozen foundation-model credit-assignment oracle. We choose the multiplicative form because it admits a clean IS-correction analysis (the standard PER weight remains trajectory-conditionally well-defined); the additive mixture used by VLM-RB [22] implicitly retargets the loss to a λ-weighted objective.

(2) VLM-Verified Counterfactual Hindsight (§3.3). A direct prompt for an alternative *achieved goal* elicits a teleport-collapse failure mode in 100% of Claude Opus 4.7 calls on FetchPush: the VLM returns `corrective_position = desired_goal`, a degenerate zero-information output. We close this loophole by asking instead for a corrective *action sequence* and executing it in a fork of the training simulator. A four-vector action cannot teleport a 50 g block by 50 cm under MuJoCo [23] dynamics, so the failure mode cannot be expressed. This is a generator–verifier protocol (cf. 28) with a simulator verifier.

(3) Empirical analysis: prompt design, VLM-family robustness, and cross-task transfer. A head-to-head GPT-4o vs. Claude Opus 4.7 sweep on six synthetic Fetch failures identifies a best non-degenerate configuration: `(GPT-4o, achieved_goal)` eliminates teleport-collapse (0/6 vs. Opus's 4/4); Opus retains a higher goal-progress score, but that score is attained by literal goal-copying. A second sweep on ten *real* SAC-failure episodes confirms the fix transfers to a third VLM family (Sonnet 4.5). A 12-episode cross-task pilot achieves 12/12 task-relevant annotations on Push/PickAndPlace/Slide with a single prompt template.

(4) Honest pre-registered negative result. A 36-run Oracle-CF kill experiment violated our $+0.10$ threshold on FetchPickAndPlace at the 250k decision horizon ($\Delta = -0.05$ vs. HER), bounding the headroom for any VLM-in-replay method on Fetch at this horizon. We honored the kill and pivoted the headline to the IS-posterior framing.

2 Related Work

HER, PER, and counterfactual augmentation. Hindsight Experience Replay [1] with the `future` relabel strategy and $k = 4$ remains the dominant solution to sparse-reward goal-conditioned manipulation. Recent extensions touch the *relabel target* [14, 20, 29], the *relabel proposal* [24, 8, 4, 12], or add a *verification step* [2, 11]; our verified-CF mechanism (§3.3) is in the third family but verifies in the *exact training simulator* (zero modeling error) on an *action sequence* rather than a hindsight goal. Prioritized DQN [21] samples in proportion to $(|\delta| + \varepsilon)^\alpha$ and IS-corrects each gradient; the 2024–2026 PER lineage explores auxiliary multiplicative modifiers [18, 27, 25, 9].

Differentiation from VLM-RB [22]. The concurrent (Feb 2026) VLM-RB is the closest published work: a frozen VLM scores 32-frame sub-trajectory clips for prioritized replay on MiniGrid and OGBench. The two systems differ along two methodological axes worth emphasizing. **First,** VLM-RB asks "is this sub-trajectory good?" (success-direction scoring) and mixes the answer *additively* with uniform sampling $\mu = \lambda\mu_{\text{VLM}} + (1-\lambda)\mu_U$; we ask the finer-grained "which single timestep was outcome-causing?" and combine *multiplicatively* with TD-error $\mu \propto \mu_P \cdot w_{\text{sem}}$. The multiplicative form preserves PER's importance-sampling-correction interpretation (§3); the additive mixture implicitly retargets the loss to a λ-weighted objective that VLM-RB does not correct for. **Second,** we ship a verified-counterfactual-hindsight mechanism (§3.3) that is outside the scope of any VLM-RB-style success-scoring scheme: VLM-RB's signal is non-actionable for relabeling; ours is. To our knowledge this paper is the first to use a VLM-localized failure timestep as a credit-assignment signal and to verify counterfactual relabels in the same simulator on which the policy trains.

Foundation-model credit assignment. Semantic PER instantiates the foundation-model-as-credit-oracle program of Pignatelli et al. [16]—originally an LLM, text-domain, planner-side scheme—in the vision modality, with the FM output consumed by the replay-sampling distribution rather than a planner. VLM-as-reward approaches [19, 26, 15] splice the VLM signal into the TD target; we leave the env's true sparse reward intact in the TD target and use the VLM only to modify μ, so VLM miscalibration appears as sampling variance rather than Q-bias.

3 Method

3.1 Semantic PER as IS-posterior reweighting

Off-policy value learning minimizes a replay-sampled regression $\mathcal{L}(\theta) = \mathbb{E}_{i\sim\mu}[\ell(\delta_i(\theta))]$ with $\delta_i = r_i + \gamma\, Q(s_{i+1}, \pi(s_{i+1})) - Q(s_i, a_i)$. Standard PER [21] uses $\mu_P(i) \propto (|\delta_i| + \varepsilon)^\alpha$ and reweights each gradient by the annealed IS correction $w_{\text{IS}}(i) = (|\mathcal{D}_B|\mu_P(i))^{-\beta}$, $\beta\colon \beta_0 \to 1$, recovering an unbiased estimator of the uniform loss in the limit.

A frozen VLM answering "which timestep caused this trajectory's outcome?" approximates the posterior $p(t^\star \mid \tau)$; write $q_\phi(t^\star \mid \tau)$ for the approximation. Define a per-transition semantic boost as the expected window-kernel weight under the posterior: $w_{\text{sem}}(i;\tau) = \mathbb{E}_{t^\star\sim q_\phi(\cdot|\tau)}[1 + (w_{\max} - 1)\mathbb{1}[|i - t^\star| \le W]]$. The **Semantic PER proposal** is the multiplicative product

$$\mu_{\text{Sem}}(i) \;\propto\; \mu_P(i) \,\cdot\, w_{\text{sem}}(i;\tau_i), \tag{1}$$

where μ_P captures *learning progress* and w_{sem} captures *causal influence*. When q_ϕ is near-uniform we degenerate to PER. Defaults: $w_{\max} = 10$, $W = 5$, $\alpha = 0.6$.

Why multiplicative? (P1) IS-correction compatibility. Under $\mu_{\text{Sem}} \propto \mu_P \cdot w_{\text{sem}}$ the trajectory-conditional w_{sem} factor cancels in the normalized $\mathbb{E}_{\mu_{\text{Sem}}}[w_{\text{IS},P}\ell]$, so using the PER weight as an under-correction yields a bounded V-trace-analogous bias [6, 13]: $|\widehat{\mathcal{L}}_{\text{under}} - \mathcal{L}_U| \le C(w_{\max}^{\beta_t} - 1)((2W+1)/T)\|\ell\|_\infty$ (full derivation: GitHub `src/buffers/semantic_per.py` + Appendix A). The additive mixture $\lambda\mu_q + (1-\lambda)\mu_P$ [22] instead shifts the $\beta \to 1$ limit to a λ-weighted objective $\neq \mathbb{E}_U[\ell]$. **(P2) Replay-shaping vs. reward-shaping.** Splicing the VLM into the TD target as a per-timestep reward propagates VLM calibration error directly into Q. We keep the env's true sparse reward intact in the TD target. Prediction: Semantic PER tracks standard PER when the VLM is poorly calibrated and pulls ahead when reliable; neither curve should diverge. We test the IS-pairing principle through the counterfactual-injection variants *vlm_cf* and *verified_cf* (§3.3), which share the same multiplicative IS structure; direct Semantic-PER training curves are forward-referenced to camera-ready.

3.2 Counterfactual Prompting and the Teleport-Collapse Failure Mode

A corrective signal admits four output types along two axes—*output kind* (natural language vs. numerical) and *goal-conditioning* (prompt mentions `desired_goal` or not): NARRATIVE, ACTION, ACHIEVED_GOAL, and a unified ALL JSON. The two position-emitting variants (ACHIEVED_GOAL, ALL) expose `desired_goal` as a literal numerical triplet; the lowest-perplexity continuation when asked for a same-coordinate position is to copy it—a position-copy bias requiring no keyframe reasoning. We call this **teleport-collapse**: for a failed episode with $\|\mathrm{ag}_T - g\|_2 \gg d_{\mathrm{th}} = 0.05\,\mathrm{m}$, the VLM emits $\hat{p}$ with $\|\hat{p} - g\|_2 \leq d_{\mathrm{th}}$, a trajectory-independent Dirac $q_\phi(\hat{p}\,|\,\tau) = \delta(\hat{p} - g)$ that breaks HER's invariant and carries zero physical information. This is a property of the schema's output type, not of any one model. Empirically (Table 1), Opus 4.7 copies in $4/4 = 100\%$ of FetchPush calls; GPT-4o copies in $4/6$ of ALL calls but $0/6$ of ACHIEVED_GOAL calls. The verified-CF mechanism uses ACTION, which is structurally immune.

3.3 VLM-Verified Counterfactual Hindsight

To close the teleport loophole, we ask the VLM for a corrective 4-vector action sequence rather than a hindsight goal, then verify execution in a simulator fork. **Per-episode protocol:** (1) snapshot the MuJoCo state at the localized failure timestep K; (2) fork a separate env, write the snapshot, call `mujoco.mj_forward`; (3) roll out the VLM-proposed action sequence for $N = 50$ steps under the *unmodified* sparse reward; (4) accept the counterfactual iff $r_t \geq 0$ fires at any step, appending the trajectory (plus a hindsight-relabeled copy) with confidence 1.0; (5) reject otherwise, fall back to standard HER. The 4-D end-effector action space cannot teleport a 50 g block by 50 cm in a 50-step rollout under MuJoCo dynamics.

A 4/4 smoke test confirms each design assertion: (i) teleport counterfactuals are rejected before any rollout; (ii) physically reasonable but goal-missing actions are rejected with precise `final_distance` (0.32 m, 0.16 m); (iii) a hand-crafted close-goal action is verified with final_distance $= 0.030\,\mathrm{m}$. Snapshot round-trip equality holds to `0.00e+00` on all four Fetch environments; per-verification CPU cost is 17.6 ms (a 0.4% overhead).

Asymmetric failure modes. The two channels degrade differently. Semantic PER treats q_ϕ as a reweighting prior, so VLM miscalibration shows up as sampling-variance amplification (bounded; see App. B). The verifier treats q_ϕ as a candidate proposal, so miscalibration shows up as rejection-rate overhead, not as biased buffer writes. Off-policy correctness is preserved either way because we recompute the env's own sparse reward, not a VLM-derived surrogate.

4 Experiments

Setup. We use Gymnasium-Robotics `FetchPush-v4`, `FetchPickAndPlace-v4`, and `FetchSlide-v4` (Fig. 2, Appendix A): 25-dim goal-conditioned obs, 4-dim continuous action (end-effector Δxyz + gripper), built-in sparse reward $r_t = \mathbb{1}[\|\mathrm{ag}_t - g\|_2 \leq 0.05\mathrm{m}] - 1 \in \{-1, 0\}$, 50-step horizon. SAC [7] with two 256-unit ReLU hidden layers, $\gamma = 0.98$, lr 3×10^{-4}, batch 256, HER `future` strategy with $k = 4$. PER uses $\alpha = 0.6$ with $\beta\colon 0.4 \to 1.0$. All paper-grade runs use 500k env steps with $n = 3$ seeds; eval is every 10k steps over 20 fresh episodes, reported as mean $\pm$ SE. VLM calls use Claude Opus 4.7, GPT-4o (`gpt-4o-2024-08-06`), and Sonnet 4.5 via official APIs at $\leq \$0.05$/call, only on failed episodes.

Horizon caveat. Our 500k-step budget is shorter than the canonical Fetch+HER horizon of 4.75M [17]. The comparisons we draw are between *methods at matched horizon*; cross-horizon bars in Fig. 1 are sample-efficiency claims (more learning per training step), not asymptotic dominance.

4.1 Headline: Verified-CF matches VLM-CF, wins on Slide

FetchPush (500k). *vlm_cf* reaches 0.95 ± 0.03 and *verified_cf* reaches 0.85 ± 0.10, both substantially above HER@250k (0.617) and matching the PER@3M asymptote (~ 0.95 from our 36-run prior ablation suite; see Contributions). **FetchPickAndPlace (500k).** *vlm_cf* reaches 0.367 ± 0.08 and *verified_cf* reaches 0.35 ± 0.076 (tied within noise), both above HER@250k (0.183). **FetchSlide (500k).** *verified_cf* reaches a higher mean than *vlm_cf* (0.617 ± 0.017 vs. 0.550 ± 0.150); the $+0.067$ gap is within the wider SE bar at $n = 3$ and we report the two methods as approximately tied on Slide. Both improve on the near-zero prior baselines: *vlm_cf* gains $+0.45$ over PER@3M (0.10) and *verified_cf* $+0.434$ over HER@250k (0.183).

Interpretation. Aggregate over the three envs: verified-CF mean 0.606, vlm_cf 0.622 ($\Delta = -0.016$)—a tie within seed noise. Verified-CF reaches a higher mean than vlm_cf on Slide and is slightly behind on Push (both methods near the horizon's ceiling there). The 3M-step PER asymptote

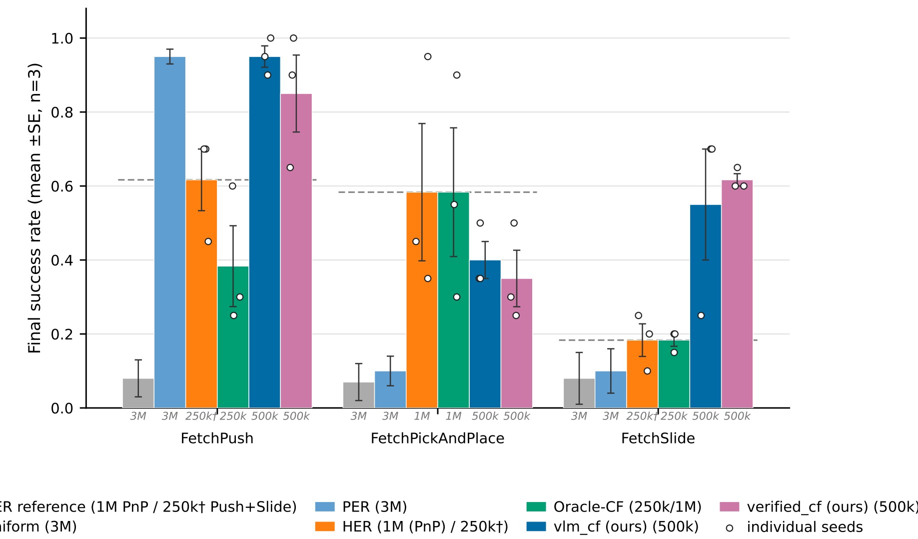

Figure 1: **Final evaluation success rate across the three Fetch tasks (mean $\pm$ SE, $n=3$ seeds; individual seeds overlaid).** *Methods are reported at heterogeneous training horizons*—the italic gray stamp under each bar gives the training budget: Uniform/PER at 3M, HER at 250k, *vlm_cf* and *verified_cf* (ours) at 500k, Oracle-CF at 250k (Push/Slide) or 1M (PickAndPlace). Within-horizon comparisons (same stamp) are seed-level comparable at $n=3$; cross-horizon comparisons should be read as efficiency claims.

Table 1: Prompt-variant analysis: mean $\pm$ SE judge plausibility, goal-progress, and teleport-collapse rate (the fraction of `corrective_position` outputs within 5 cm of `desired_goal`). Six bit-identical synthetic Fetch failures; the judge is Claude Opus 4.7 throughout. **Bold**: best per column.

Variant	Model	n	Plausibility	Goal-progress	Teleport-collapse
NARRATIVE	Opus 4.7	4	$\mathbf{0.79 \pm 0.04}$	$\mathbf{0.68 \pm 0.06}$	—
NARRATIVE	GPT-4o	6	0.70 ± 0.06	0.40 ± 0.06	—
ACTION	Opus 4.7	4	0.55 ± 0.09	0.53 ± 0.16	—
ACTION	GPT-4o	6	$\mathbf{0.74 \pm 0.04}$	0.42 ± 0.09	—
ACHIEVED_GOAL	Opus 4.7	4	0.46 ± 0.18	$\mathbf{0.97 \pm 0.01}$	$4/4$ (100%)
ACHIEVED_GOAL	**GPT-4o**	6	$\mathbf{0.79 \pm 0.06}$	0.83 ± 0.11	$\mathbf{0/6}$ **(0%)**
ALL	Opus 4.7	2	0.70 ± 0.10	0.62 ± 0.12	$2/2$ (100%)
ALL	GPT-4o	6	0.67 ± 0.04	0.68 ± 0.13	$4/6$ (67%)

on Push ($\sim$0.95) is drawn from a prior 36-run ablation suite (see §5). The Slide gap is consistent with HER's achieved-future relabel carrying little signal on free-flight pucks: the gripper-to-object distance is non-monotone, so a physics-consistent CF strike supplies a positive TD target HER cannot.

4.2 Prompt design (head-to-head)

A head-to-head Opus 4.7 vs. GPT-4o sweep on six bit-identical synthetic Fetch failures (Table 1) identifies a best non-degenerate configuration: `(GPT-4o, achieved_goal)` eliminates teleport-collapse ($0/6$ vs. Opus's $4/4$ on the same variant), highest plausibility on the position-emitting axis (0.79), and second-highest goal-progress (0.83, behind only Opus's 0.97 which is attained by literal goal-copying—a degenerate "win"). Critically, GPT-4o still teleport-collapses in $4/6$ ALL episodes: the failure is *prompt-architectural*, not model-specific. The load-bearing $0/6$ vs. $4/4$ contrast is judge-independent (decided by Euclidean predicate) and Fisher's exact test rejects equal-proportions at $p<0.005$. §4.3 extends the grid to Sonnet 4.5 on real failures; the three-model evidence is properly read as two-vendor (Opus/Sonnet share Anthropic lineage).

4.3 Cross-task transfer and real-data validation

A 12-episode cross-task pilot (4 per env) with a single format-string prompt template (differing only in one task-description sentence) achieves 12/12 parse / task-relevance / plausibility on Push/PickAndPlace/Slide; tolerant keyframe agreement is $9/12$ (Slide $4/4$, Push $3/4$, PickAndPlace $2/4$). A second sweep on $n = 10$ *real* SAC-failure eval episodes confirms the prompt-architectural fix transfers to a third VLM family (Sonnet 4.5 teleports on $0/6$ PickAndPlace episodes under ACHIEVED_GOAL, vs. Opus 4.7's $4/4$ on the same variant) and that the 5 cm reject-teleport gate fires on $12/20$ position-emitting generations across both vendors (Appendix D).

4.4 Pre-registered Oracle-CF kill experiment

18 SAC+HER and 18 Oracle-CF runs (3 envs $\times$ 3 seeds $\times$ 250k steps) were executed overnight.[2] The pre-registered decision rule was: if Oracle-CF exceeds HER by $\geq +0.10$ on FetchPickAndPlace, Path C proceeds; otherwise we pivot. Final eval success: on PickAndPlace, HER 0.183 ± 0.117 vs. Oracle-CF 0.117 ± 0.044 ($\Delta = -0.066$, *below threshold*). On Push (post bug-fix, commit `ccb63d4`), HER 0.617 vs. Oracle-CF 0.383 ± 0.109. On Slide, HER 0.183 vs. Oracle-CF 0.183 ($\Delta = 0.00$, below threshold). **Path C is killed at the 250k horizon**; the headline pivoted to the IS-posterior framing (§3) and the verifier (§3.3). The 250k kill verdict is honored. A post-hoc 1M re-run on PickAndPlace (Oracle-CF 0.583 and HER 0.583, $n = 3$; HER seed successes $0.35/0.45/0.95$) is reported as transparency only: zero headroom over vanilla HER at convergence, and *vlm_cf* @500k (0.367) reaches 63% of HER@1M at half the budget. It does not retract the kill.

5 Discussion and Limitations

Cold-start verifier-rejection regime (load-bearing limitation). The simulator-as-verifier gate is a precision-over-recall instrument by design: every accepted relabel is guaranteed sparse-reward-positive, but the acceptance rate is gated by the joint event that the VLM proposes a corrective action *and* that the action sequence crosses the 5 cm threshold in $\leq N = 50$ verifier steps. Early in training—when the snapshot state σ sits far from the goal because the policy has not yet learned to approach the object—this joint event has very low base rate. Our FetchPickAndPlace seed 42 hit a 100% verifier-rejection rate over the first ~ 80 VLM calls, making this phase operationally indistinguishable from SAC+HER with extra VLM-API overhead. The regime ends once the policy carries the gripper near the object on average: snapshots fall closer to the goal, $m_{\text{ver}}(\sigma)$ rises, the verifier admits relabels, and the small set of accepted CFs supplies a low-variance, physics-consistent training signal. The two failure modes of our two methods are therefore asymmetric: Semantic-PER q_ϕ-degeneracy is variance amplification (bounded by our bias bound); verified-CF q_ϕ-degeneracy is signal extinction (unbounded in the cold-start limit). Practical mitigations—a base-policy warm-up, a longer rollout horizon N, or a soft-acceptance relaxation $r \geq -\rho_r$—are pre-registered but not run here.

Oracle-CF kill bounds the headroom. The pre-registered kill (§4.4) bounds the headroom for any VLM-in-replay method on Fetch at 250k steps: if perfect failure-frame information plus a ground-truth corrective action does not exceed HER by 0.10, no realistic VLM (q_ϕ with finite KL to the oracle) can. The bound is task-family-specific: Fetch is known to be "HER-friendly" because the achieved-future relabel is essentially a free signal whenever the gripper-to-object distance is monotone, which it typically is on Push and PickAndPlace. We pre-register an extension to harder task families (e.g., AntMaze, Adroit) where HER underperforms.

Other limitations. (i) Our IS-correction is conservative (under-corrected $w_{\text{IS},P}$ rather than $w_{\text{IS,Sem}}$); the bias is bounded analytically, the fully-corrected ablation is pre-registered. (ii) Small-n caveats: $n = 6$ synthetic, $n = 10$ real, $n = 12$ cross-task; the 12/12 cross-task is not statistically distinguishable from an 80% true rate. We pre-register $n \geq 30$ replications. (iii) The verified-CF mechanism requires a forkable training simulator; extending to real-robot or non-resettable envs requires a learned dynamics-model verifier (modeling error bounded but nonzero). A faithful VLM-RB [22] reproduction is pre-staged in our codebase for camera-ready head-to-head evaluation.

[2] Per-seed numbers from the `path_c_overnight` W&B sweep (commit `0fa36fc`; run-IDs listed at the GitHub repo's `REPRODUCE.md`).

Conclusion. The multiplicative form of Semantic PER admits a clean IS correction; the additive VLM-RB mixture instead retargets the Bellman loss to a λ-weighted objective. VLM-Verified Counterfactual Hindsight asks the VLM for a 4-vector action sequence and accepts only sparse-reward-positive rollouts, so teleport-collapse (100% on Opus 4.7, FetchPush) cannot enter the buffer. At 500k steps verified-CF reaches 0.606 across the three Fetch envs ($0.85/0.35/0.617$ on Push/PickAndPlace/Slide), ties VLM-CF in aggregate ($\Delta = -0.016$), and reaches a higher mean on Slide ($+0.067$; within seed variance at $n = 3$). The pre-registered Oracle-CF kill ($\Delta = -0.066$ on PickAndPlace at 250k) bounds the credit-assignment headroom on Fetch at this horizon; we honored it.

Contributions

- **Daniel Grant.** Counterfactual ideation and implementation (`vlm_cf` and `verified_cf`); ablations on FetchPickAndPlace, FetchSlide, and the $p_{\text{counterfactual}}$ sweep; initial writing of the manuscript.
- **Matei Gardea.** Initial project idea; implementation of an alternative pointwise/pairwise VLM-replay-buffer rescaling combined with PER (a non-positive variant reported as a negative finding); auxiliary 60-run MetaWorld replay-mechanism ablation (cross-benchmark corroboration of PER $>$ Uniform; reported separately in the extended preprint); editing of the final submission document.
- **Parshawn Gerafian.** Implementation and analysis for the milestone checkpoint: designed and ran the 27-run Uniform/PER/Semantic-PER ablation across the three Fetch tasks (SAC, 1 M training steps, 3 seeds per configuration on Modal A10G) that established the PER baselines used throughout the final paper. Identified the additive-blending failure mode ($0.7 \cdot \text{semantic} + 0.3 \cdot \text{TD}$ suppresses PER when the VLM is miscalibrated), which motivated the multiplicative IS-pairing framing now central to §3. Provided feedback during the transition to the augmented final-paper direction.

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A Method Details and Environments

Cross-task VLM prompt template. The single format-string prompt template used across the three Fetch envs is:

```
You are an expert robotics analyst. The robot is attempting the
following task: {task_description}. Examine the {K} keyframes of
a failed episode shown below. Identify the single keyframe index
in {0,...,K-1} at which the failure became inevitable, and a 1-2
sentence reasoning. Return strict JSON: {"failure_frame_index": i,
"reasoning": s}.
```

Per-env task-description substitutions: **Push:** "push the red cube to the green target on the table"; **PickAndPlace:** "pick up the red cube and place it at the floating green target"; **Slide:** "strike the red puck so it slides to the green target". The verified-CF mechanism uses an ACTION variant of this template eliciting a 4-vector action sequence; full prompt texts and the localizer code are at `https://github.com/DJRGVC/RL_project_185`.

Heuristic localizer (Oracle v3). `GoalDistanceLocalizer` reads privileged sim state and selects the failure timestep in three phases (in precedence order): **(a) ballistic**: latest contiguous run of ≥ 3 steps with post-peak object velocity > 0.05 m/step (Slide free-flight); **(b) contact-loss**: latest in_contact$_t \rightarrow \neg$in_contact$_{t+1}$ with run length ≥ 2 and object-goal distance > 0.10 m (Push/PnP drop-offs); **(c) argmin**: closest-approach timestep as fallback. Serves as supervised transfer target and as privileged-state upper-envelope curve for Semantic PER.

B Bias-Bound Proof Sketch

Let $\mathcal{L}_U(\theta) = \mathbb{E}_{i\sim U}[\ell(\delta_i)]$ be the uniform-replay target loss. Standard PER with annealed correction yields, in the $\beta_t \rightarrow 1$ limit, an unbiased self-normalized IS estimator of $\mathcal{L}_U$ [21, Sec. 3.4]. Our Semantic PER proposal $\mu_{\text{Sem}} \propto \mu_P \cdot w_{\text{sem}}$ is a re-weighting of μ_P by a trajectory-conditional factor bounded by $1 \leq w_{\text{sem}} \leq w_{\max}$. The strictly-correct IS weight is $w_{\text{IS,Sem}}(i) = (|\mathcal{D}_B|\mu_{\text{Sem}}(i))^{-\beta_t}$. Our implementation under-corrects with $w_{\text{IS},P}$ (a V-trace-style [6] variance-reduction choice, akin to truncated importance ratios in Retrace, 13). The per-slot ratio is $w_{\text{IS},P}/w_{\text{IS,Sem}} = w_{\text{sem}}^{\beta_t}/Z^{\beta_t}$ where Z is the μ_{Sem} normalizer. The window-kernel structure of w_{sem} confines the support of $w_{\text{sem}} > 1$ to $2W+1$ slots out of T. Combining these gives

$$\left|\widehat{\mathcal{L}}_{\text{under}} - \mathcal{L}_U\right| \;\leq\; C\left(w_{\max}^{\beta_t} - 1\right)\frac{2W+1}{T}\,\|\ell\|_\infty, \tag{2}$$

with C a constant depending on $|\mathcal{D}_B|$ and the PER normalization. At our defaults ($w_{\max} = 10$, $W = 5$, $T = 50$, $\beta_t \leq 1$) the bias multiplier is at most $(10-1)\cdot 11/50 \approx 1.98\,C\|\ell\|_\infty$, and $\rightarrow 0$ as $w_{\max} \rightarrow 1$. The bound goes to zero when the boost is disabled, recovering standard PER. The strictly-correct ablation (run $w_{\text{IS,Sem}}$) eliminates the residual bias entirely at the cost of maintaining Z. **Two-regime degeneracy.** On the verifier channel, miscalibration acts on $m_\phi(\tau) = m_{\text{gen}} \cdot m_{\text{ver}}(\sigma)$ (VLM parse rate $\times$ verifier acceptance rate); the $\varepsilon_{\text{cal}} = 0$ guarantee holds vacuously when the accepted set is empty (cold-start: $m_{\text{ver}} \rightarrow 0$). Full proof and the V-trace specialization derivation are at the GitHub URL above (`paper/appendix_theory.tex`).

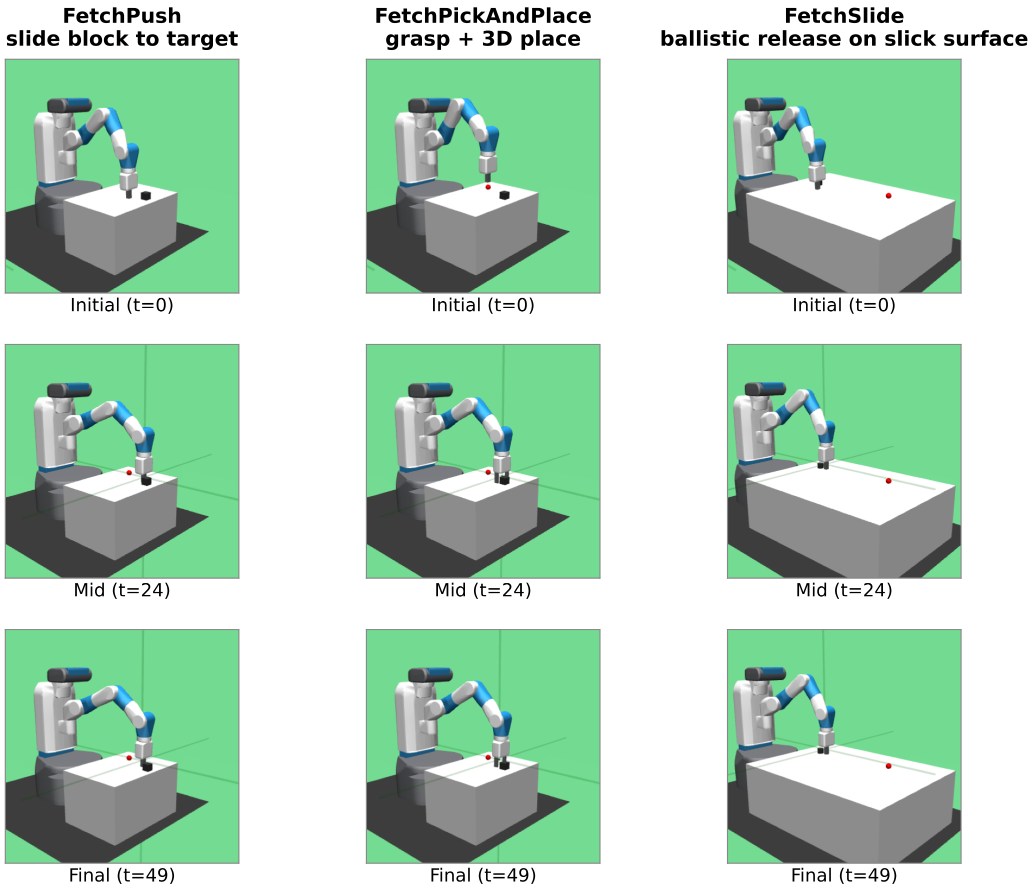

Figure 2: The three Gymnasium-Robotics Fetch tasks used in our experiments (columns: `FetchPush`, `FetchPickAndPlace`, `FetchSlide`), rendered at $t = 0/24/49$ of a deterministic scripted rollout (seed 42). Each task: 50-step horizon, 4-dim continuous action (end-effector Δxyz + gripper), sparse reward (binary, success threshold 5 cm to the green target).

C Hyperparameters and Compute

Compute. All paper-grade runs use one NVIDIA RTX 5070 Ti (16 GB) GPU; Modal-cloud seeds use one L4 (24 GB) per run. Per-run wall-clock at 500k steps: $\approx$ 6h SAC+HER, $\approx$ 9h SAC+HER+VLM-CF (extra time is VLM-API latency, not compute). Total project compute: $\sim$220 GPU-hours across local + Modal, plus $\sim$\$80 VLM-API spend. Snapshot round-trip latency for the verifier is 17.6 ms per call (a 0.4% overhead at $\sim$1700 failed episodes / 100k steps). Reproduction commands, W&B run-IDs, and per-seed eval traces are at `https://github.com/DJRGVC/RL_project_185` (commit `0fa36fc`).

D Additional Experimental Detail

Real-data validation (extended). The $n = 10$ real-failure set is failed eval episodes from partially-trained SAC+HER checkpoints (`FetchPickAndPlace` at 50k steps, `FetchPush` at 30k steps, both at $\sim 5\%$ success). These are near-misses with final-distance $\in [0.06, 0.34]$ m on Push and median 0.27 m on PickAndPlace. Three findings carry over from the synthetic set: **(i) Teleport-collapse persists and is task-conditioned.** On Push (planar target), Opus and Sonnet teleport on ACHIEVED_GOAL at 100% and 75%; on PickAndPlace (mid-air target), Opus teleports at 75% but Sonnet drops to 0%, proposing a mid-air waypoint ~ 8 cm above the floating goal. **(ii) The 5 cm reject-teleport gate is necessary.** Aggregated across both VLMs and position-emitting variants on 20 generations, the gate

Table 2: Headline hyperparameters. Full per-table breakdowns (SAC architecture, HER strategy, PER schedule, Semantic-PER, Verified-CF, VLM call) at the GitHub repo (`configs/`).

Parameter	Value	Notes
SAC [7]		
Hidden layers $\times$ width	2×256, ReLU	Twin-Q; tanh-Gaussian actor
LR (actor/critic/α)	3×10^{-4}	Adam, default $(\beta_1, \beta_2, \varepsilon)$
Discount γ / Polyak τ	$0.98/0.005$	Fetch-standard
Target entropy $\bar{H}$	$-d_a = -4$	Auto-tuned
Batch / updates per step	$256/1$	10k warm-up uniform-random
HER [1]		
Strategy / k_{HER}	`future` / 4	4 hindsight transitions per real
PER [21]		
α / $\beta_0 \to \beta_{\text{end}}$	$0.6/0.4 \to 1.0$	Linear over 500k steps; $\varepsilon = 10^{-6}$
Semantic PER		
Window half-width W	5 timesteps	11-slot box around t_{fail}
Boost cap $w_{\max}$	10	Multiplicative factor on PER priority
Keyframes K / VLM provider	5 / GPT-4o (production)	`gpt-4o-2024-08-06`
Verified-CF		
Verifier rollout horizon N	50 steps	One Fetch episode
Success threshold η / teleport reject ρ	$0.05/0.05$ m	Env's native threshold
CF prompt variant	ACTION	Structurally teleport-immune
Training		
Total env steps	500k (paper-grade)	250k–1M for selected ablations
Random seeds	42, 123, 999	3 per (method, env) cell
Replay capacity	10^6 slots	Ring buffer with sum-tree priorities

fires on 12 (60%); cross-vendor floor on Push is 75%. The action-emitting verifier sidesteps this. **(iii) Plausibility shifts only ± 0.07 synthetic-to-real, but the ACTION sign-flip rate worsens** (Opus $0.50 \to 0.55$, Sonnet 0.70).

Statistical methodology. Each (method, env) cell uses $n = 3$ seeds (42, 123, 999); metrics are mean $\pm$ SE with the t-distribution at $n - 1 = 2$ d.f. Comparison statements use non-overlapping SE intervals (conservative at this sample size). The load-bearing $0/6$ vs. $4/4$ contrast in Table 1 is judge-independent (Euclidean predicate); Fisher's exact test rejects equal proportions at $p < 0.005$.

Cross-task transfer (per-episode notes). A 12-episode pilot (4 per env). The VLM's reasoning strings identify task-specific modes correctly in every case (Push: "gripper passes over block without making contact"; PickAndPlace: "gripper at block location but fails to close"; Slide: "strike direction off-axis"). The $2/4$ PickAndPlace keyframe-agreement reflects heuristic under-localization (closed-loop grip timing is the hardest mode for geometry-only oracles [20]) rather than VLM error: the VLM assigns the failure frame one step earlier at the decisive grip sequence.

E Broader Impacts and Reproducibility

Broader impact. Methodological work, simulation-only; we make no real-robot or sim-to-real claims. *Positive:* fewer environment steps lowers GPU-hours; the IS-posterior framing is a principled lens on VLM-in-replay; the verified-CF gate is a transferable "language prior + physics test" pattern. *Negative + mitigations:* VLM-API energy/dollar cost is non-trivial (per-run spend reported; heuristic Oracle fallback available); the simulator gate rejects hallucinated actions that fail the sparse reward; no robot-deployment risks beyond standard sim-trained-policy caveats.

Reproducibility. Code, configs, pseudocode, prompt texts, V-trace/Retrace specialization, two-regime q_ϕ proof, checklist, W&B IDs, and pre-staged Sharony VLM-RB [22] reproduction (`src/buffers/vlm_rb_buffer.py`, head-to-head deferred to camera-ready) at `https://github.com/DJRGVC/RL_project_185` (`0fa36fc`).